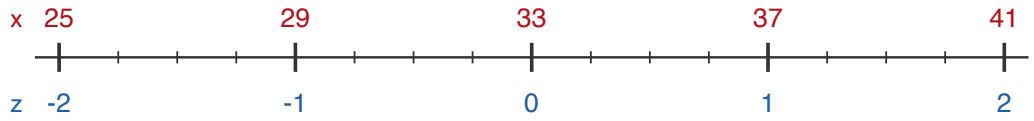
Answers: Z-Values for Commute Times

The commute times of students to campus have mean $\mu = 33$ minutes and standard deviation $\sigma = 4$ minutes.

The z-value of a commute time x is

$$z = \frac{x - \mu}{\sigma}$$

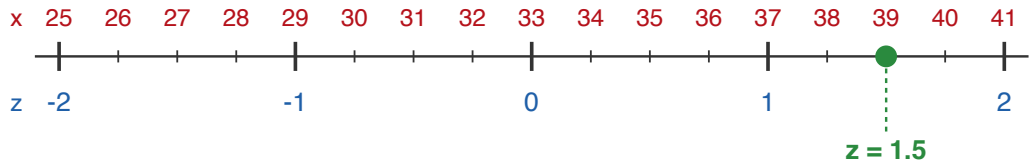
a) Fill in the x-values for each z-value on the number line.



$\mu = 33$ is at $z = 0$. Each step of one standard deviation adds or subtracts 4:

- $33 + 4 = 37$ and $37 + 4 = 41$
- $33 - 4 = 29$ and $29 - 4 = 25$

b) Ana's commute is $x = 39$ minutes

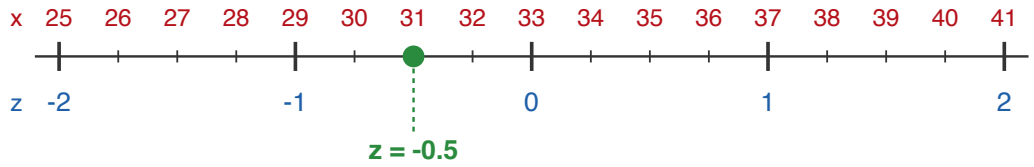


Visually: 39 is 6 minutes above the mean. Each tick is 1 minute and 4 ticks make one standard deviation, so 6 minutes is $6/4 = 1.5$ standard deviations above the mean: $z = 1.5$.

Calculate:

$$z = \frac{39 - 33}{4} = \frac{6}{4} = 1.5$$

c) Ben's commute is $x = 31$ minutes

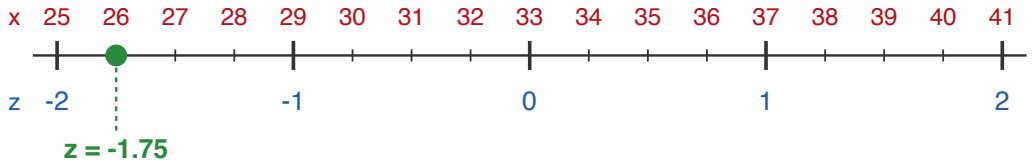


Visually: 31 is 2 minutes below the mean. Each tick is 1 minute and 4 ticks make one standard deviation, so 2 minutes is $2/4 = 0.5$ standard deviations below the mean: $z = -0.5$.

Calculate:

$$z = \frac{31 - 33}{4} = \frac{-2}{4} = -0.5$$

d) Carla's commute is $x = 26$ minutes

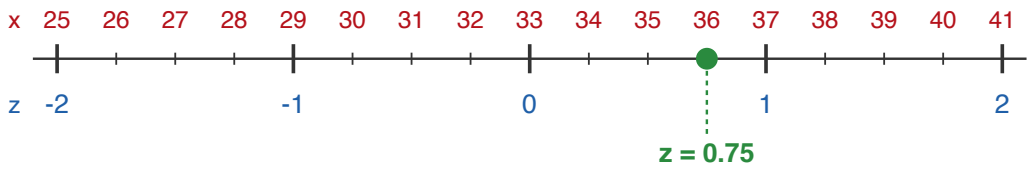


Visually: 26 is 7 minutes below the mean. Each tick is 1 minute and 4 ticks make one standard deviation, so 7 minutes is $7/4 = 1.75$ standard deviations below the mean: $z = -1.75$.

Calculate:

$$z = \frac{26 - 33}{4} = \frac{-7}{4} = -1.75$$

e) Dev's commute is $x = 36$ minutes



Visually: 36 is 3 minutes above the mean. Each tick is 1 minute and 4 ticks make one standard deviation, so 3 minutes is $3/4 = 0.75$ standard deviations above the mean: $z = 0.75$.

Calculate:

$$z = \frac{36 - 33}{4} = \frac{3}{4} = 0.75$$